

Solution Manual

Eric, Justin, Yichen

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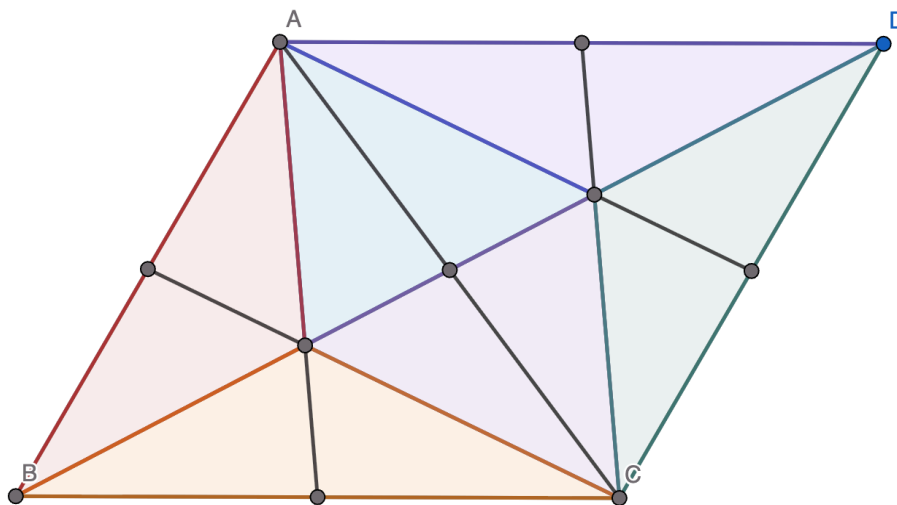
Eric

Q1 Solution Geometry

In $\triangle ABC$, the A -median, B -median, C -median have lengths 13, 14, 15, respectively. Find area of $\triangle ABC$.

Answer: 112

Solution: Notice that the area of the triangle with side lengths 13, 14, and 15 has area $\sqrt{21(21-13)(21-14)(21-15)} = 84$. Let S be the area of $\triangle ABC$. Consider reflecting B over the midpoint of AC to obtain D such that quadrilateral $ABCD$ is a parallelogram. Thus, due to the fact that the centroid is exactly two-thirds of the way in each median, $2S = 6 \cdot \left(\frac{2}{3}\right)^2 \cdot 84 \Rightarrow S = \frac{4}{3} \cdot 84 = 112$.



Q2 Solution Number Theory

Given a positive base-10 positive integer $N = a_n a_{n-1} \dots a_1 a_0$, let $f(N) = 2^n a_0 + 2^{n-1} a_1 + \dots + 2^1 a_{n-1} + 2^0 a_n$. Find $f(f(\dots f(2014^{2014}) \dots))$ (2014^{2014} f 's).

Answer: 19

Solution: Firstly, notice that if N is a positive integer, then $f(N) > 0$. Also, if $N \geq 100$, then $n \geq 2$, $f(N) \leq 9 \cdot 2^n + 9 \cdot 2^{n-1} + \dots + 9 < 9 \cdot 2^{n+1} < 10^n \leq N$. Also, if $20 \leq N \leq 100$, then if $N = a_1 a_0$, then $a_1 \geq 2$ and thus $f(N) = 2a_0 + a_1 < 10a_1 + a_0$. Thus, $f(N) < N$ whenever $N > 19$. Thus, because there are 2014^{2014} applications of f onto 2014^{2014} , the final result $f(f(\dots f(2014^{2014}) \dots))$ (2014^{2014} f 's) must be at most 19 and at least 1. Now, I claim that if N is divisible by 19, then $f(N)$ is also divisible by 19. Notice that $10^n f(N) = 10^n (2^n a_0 + 2^{n-1} a_1 + \dots + 2^1 a_{n-1} + 2^0 a_n) = 10^0 \cdot 20^n a_0 + 10^1 \cdot 20^{n-1} a_1 + \dots + 10^n \cdot 20^0 a_n \equiv 10^n a_n + 10^{n-1} a_{n-1} + \dots + 10^0 a_0 = N \pmod{19}$. This proves the claim. Notice that 2014^{2014} is divisible by 19. Thus, $f(f(\dots f(2014^{2014}) \dots))$ (2014^{2014} f 's) is divisible by 19. Also because that the previous expression is at most 19 and at least 1, thus $f(f(\dots f(2014^{2014}) \dots))$ (2014^{2014} f 's) must be exactly 19.

Yichen

Q1 Solution

Rotate triangle APC 60 degrees counterclockwise to triangle AP'C' such that C goes to B. Then, in triangle APP', $\angle P'AP = 60$, $AP = AP'$, so $PP' = 6$ too. Then, PP'B is a 30-60-90 triangle. Then,

$\angle APB = 120$, and by Law of Cosines we can find that $AB = 6\sqrt{7}$.

Q2 Solution

Consider the complementary. If Kait rolls a 6 on the i th turn, then the next i and the previous $i - 1$ rolls are all not 6. The probability of this

happening is $\left(\frac{5}{6}\right)^{2i-1} \cdot \left(\frac{1}{6}\right)$. The sum of all expressions is a

geometric sequence with initial term $\frac{5}{36}$ and common ratio $\frac{25}{36}$. Thus,

the probability the next N rolls doesn't contain a 6 is $\frac{5}{11}$, so the answer

is $\frac{6}{11}$.

Justin

Q1 Solution (Geometry)

Solution. Let E be the foot of the altitude from B to AC . Then we have

$$\begin{aligned}4 &= AH \\&= \frac{AE}{\sin \angle AHE} \\&= \frac{AB \cos \angle BAC}{\sin \angle BCA} \\&= \frac{AB \cos \angle BAC}{\frac{AB}{2R}} \\&= 2R \cos \angle BAC,\end{aligned}\tag{1}$$

where R is the circumradius of $\triangle ABC$.

Note that $\cos \frac{\angle BAC}{2} = \cos \angle YAC = \frac{\sqrt{21}}{6}$ from the Pythagorean theorem. Using the double-angle formula for cosines, we have

$$\begin{aligned}\cos \angle BAC &= 2 \cos^2 \frac{\angle BAC}{2} - 1 \\&= 2 \left(\frac{\sqrt{21}}{6} \right)^2 - 1 \\&= \frac{1}{6}.\end{aligned}\tag{2}$$

Substituting (2) into equation (1) gives $R = 12$. Note that (2) also gives us $\sin \angle BAC = \frac{\sqrt{35}}{6}$ because $0^\circ < \angle BAC < 180^\circ$.

By the extended Law of Sines,

$$\begin{aligned}BC &= 2R \sin \angle BAC \\&= 2 \cdot 12 \cdot \frac{\sqrt{35}}{6} \\&= \boxed{4\sqrt{35}}.\end{aligned}\quad \square$$

Q2 Solution (Number Theory)

Solution. Suppose some prime number p divides $20^{22} + 1$. Then we have

$$20^{22} \equiv -1 \pmod{p}$$

$$20^{44} \equiv 1 \pmod{p}$$

Let d be the order of $20 \pmod{p}$. It follows that $d \nmid 22$ and $d \mid 44$, so either $d = 44$ or $d = 4$.

Case 1: $d = 44$. We have $20^{p-1} \equiv 1 \pmod{p}$ from Fermat's little theorem. Since d is the order of $20 \pmod{p}$, we must have $44 \mid p - 1$. The smallest such p is 89, so we check to see if $p = 89$ works:

$$2^{44} \equiv 2048^4 \pmod{89}$$

$$\equiv 1^4 \pmod{89}$$

$$\equiv 1 \pmod{89}$$

$$5^{22} \equiv 25 \cdot (625)^5 \pmod{89}$$

$$\equiv 25 \cdot 2^5 \pmod{89}$$

$$\equiv -1 \pmod{89}$$

Multiplying these tells us that $20^{22} \equiv -1 \pmod{89}$, as desired.

Case 2: $d = 4$. This means that $20^4 \equiv 1 \pmod{p}$, or equivalently $3 \cdot 7 \cdot 19 \cdot 401 \equiv 0 \pmod{p}$. It can be checked that none of 3, 7, 19 divide $20^{22} + 1$. However,

$$20^{22} \equiv 400^{11} \pmod{401}$$

$$\equiv (-1)^{11} \pmod{401}$$

$$= -1 \pmod{401},$$

so $p = 401$ works.

Conclusion. Since we have shown that $p = 89$ and $p = 401$ work, it remains to prove that no more p with $p < 401$ and $44 \mid p - 1$ divide $20^{22} + 1$. The only p satisfying these conditions are 353 and 397, so we wish to show that $20^{22} \not\equiv -1 \pmod{\{353, 397\}}$.

Let us check 397 first. We have

$$\begin{aligned} 20^{22} &\equiv 3^{11} \pmod{397} \\ &\not\equiv -1 \pmod{397}. \end{aligned}$$

Now we just have to check 353. Note that

$$\begin{aligned} 2^{44} &\equiv 2^4 \cdot 1024^4 \pmod{353} \\ &\equiv 2^4 \cdot (-35)^4 \pmod{353} \\ &\equiv 70^4 \pmod{353} \\ &\equiv -1 \pmod{353}. \end{aligned}$$

Hence it remains to show that $5^{22} \not\equiv 1 \pmod{353}$:

$$\begin{aligned} 5^{22} &\equiv 25 \cdot 625^4 \pmod{353} \\ &\equiv 25 \cdot 81^4 \pmod{353} \\ &\equiv 25 \cdot 207^2 \pmod{353} \\ &\equiv 233 \cdot 207 \pmod{353} \\ &\not\equiv 1 \pmod{353} \end{aligned}$$

We have shown that neither 353 nor 397 divide $20^{22} + 1$, so our final answer is $89 + 401 = \boxed{490}$. □